

Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

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Abstract: Ecological networks are increasingly used to understand biodiversity and predict ecosystem responses to environmental change, yet their application is often limited by uncertainty about the assumptions embedded in different network representations. Here, we present a theory-driven framework that views network construction as a hierarchical transition from metawebs, which represent the feasibility of interactions, to realised webs, which represent interactions expressed in specific spatiotemporal contexts. We identify five key processes underlying this transition: evolutionary compatibility and co-occurrence, which define interaction feasibility, and abundance, diet choice, and non-trophic interactions, which determine interaction realisation. We further position these processes along a continuum of network construction methods, from inductive approaches that infer structure from observations to deductive approaches that generate structure from mechanistic principles. By explicitly linking network representations to their underlying assumptions and scale-dependent processes, our framework clarifies the questions that different networks can address, highlights key challenges in moving between potential and realised interactions, and provides a roadmap for developing predictive network ecology. We argue that greater conceptual clarity is essential for advancing biodiversity science and for the effective application of networks in conservation, environmental management, and forecasting ecological change in the Anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.*, foundational studies on food web structure and robustness: [5–8]). Documenting interactions is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5,10,11]; recent discussions: [12,13]). Against this, it is important to evaluate the value and the limitations of the various network conceptualisations and how these relate to biodiversity concepts, such as community structure or ecosystem function [14]. In this perspective we aim to provide an overview of different **food web** representations, particularly how each representation embeds assumptions about the processes that determine interactions (Section 3) about the levels of organisation at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales) and the way in which we construct the resulting networks (Section 4).

Network construction reflects both the data used and the theories governing species interactions. We still lack a clear explanation of the different assumptions and scale dependent processes that underpin network construction alongside extensive discussions about the challenges relating to data collection and observation [6,10,14–20]. Such an understanding should deliver an acceleration in capacity to more effectively predict the impact of multiple stressors on biodiverse communities.

In their recent work, [21] synthesised ecological networks based on node resolution and link type, with a focus on the data and methodologies underlying each representation. Here, we take a complementary theory-driven perspective by framing network construction as a hierarchical transition from feasibility to realisation of interactions. In the following sections, we review how nodes, edges, and interaction processes shape different network representations, highlighting how changes in assumptions alter ecological scale, node and

link resolution, modelling approaches, and the relative importance of evolutionary versus ecological processes. We conclude by aligning these representations with key questions in biodiversity science in the Anthropocene.

2 Setting the Scene: The Not So Basics of Nodes and Edges

Ecological networks serve multiple uses, representing an ‘object’ from which inferences can be made. While many aspects of community structure can be analysed without networks (*e.g.*, trait distributions or abundance patterns) networks provide a formal framework for capturing the organisation among species. The study of network structure and topology has a long history in ecology, rooted in early theory on energy flow [2,22], and later extended to questions of robustness, stability, and complexity (*e.g.*, [7,20,23–25]). More recent work has built on this foundation to link network structure to ecosystem functioning, persistence, and dynamical behaviour (*e.g.*, [26–28]). Networks are therefore commonly treated as response variables in tests of ecological theory and statistical models of the generative processes that give rise to observed structure and are widely used to compare communities across environmental gradients or through time [*e.g.*, 29,30]. They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss (*e.g.*, [7,31–33]), as well as inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades (*e.g.*, [34–36]). Against this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the levels of organisation at which they are defined, take many forms [17,37], each of which encode a series of assumptions within a network. Here we introduce a perspective on these baseline assumptions.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.*, [38,39]), feeding guilds (*e.g.*, [40]), or life-stage-specific subsets of species (*e.g.*, [41]). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation per se. Representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.*, whether species *a* consumes species *b*), bias estimates of degree distributions (particularly

generality and vulnerability) and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured [41,42]. At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds [43]. More broadly, issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.*, [6,44]).

2.2 What is captured by an edge?

Understanding edges requires distinguishing between *potential* and *realised* links - potential links reflect feasibility, whereas realised links reflect fluxes such as energy transfer. Links within food webs are thus a representation of either potential links between species [45] or fluxes within a system *e.g.*, energy transfer or material flow as the result of the feeding links between species [2,46]. Edges can thus correspond to different ‘currencies’ [21]. There is also a myriad of ways in which the links themselves can be specified. Links between species can be treated as present or absent (*i.e.*, binary), may be defined as probabilities [47,48] or by continuous functions which further quantify the strength of an interaction [49]. Link definition depends on both the ecological currency and how interactions are represented. For example, feasibility is unlikely to accommodate flux, but does align with binary or probability representations. Taking a food web that consists of links representing feasible interactions among a collection of species will be meaningless if one is interested in understanding the flow of energy through the network as the links are not environmentally/energetically constrained.

2.3 Network representations

Given these definitions of nodes and edges, ecological networks can be divided into two broad types: **metawebs**, which represent all *potential* interactions within a species pool [44], and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. These representations differ in both the scales at which they are constructed and the processes assumed to generate their structure.

A metaweb is fundamentally a list of *feasible* interactions between species. Feasibility is determined by trait complementarity, typically related to feeding, and can be further

96 constrained by species co-occurrence, producing a transition from *global* to *regional*
97 metawebs. Metawebs therefore identify evolutionarily and regionally plausible interactions,
98 ecologically impossible (*i.e.*, forbidden) links [50], and the potential diet breadth of species
99 [51].

100 In contrast, realised networks are localised in space and time, with links shaped by co-
101 occurrence, environmental conditions, and diet choice. Even when represented as binary
102 matrices, their links implicitly reflect interaction strength and energy transfer. Realised
103 networks are therefore not simple spatial or temporal subsets of metawebs; they emerge
104 from the processes governing whether feasible interactions are expressed. Consequently,
105 a metaweb and realised network containing the same species may differ substantially in
106 structure because link presence is governed by different constraints. Links absent from a
107 metaweb represent infeasible interactions, whereas links absent from a realised network
108 reflect context-dependent ecological constraints.

109 A key distinction between these representations concerns what constitutes a realised
110 interaction. At its most general, a realised network requires more than the feasibility of
111 species interactions; it must incorporate the ecological context that determines whether
112 feasible interactions are expressed. This context may be represented through observation,
113 abundance-dependent encounter rates, consumer choice, environmental conditions, or
114 other mechanisms that constrain interaction expression. Consequently, realised networks
115 need not be observed directly, but they should reflect processes governing the distribution
116 or strength of interactions within a particular community. In contrast, metawebs represent
117 the space of feasible interactions irrespective of whether those interactions are expressed
118 locally.

119 **3 From Nodes and Edges to Process and Constraints**

120 In the previous section we discussed how the definition of nodes and edges, representing
121 different scales and processes, lead to the concept of a metaweb and a realised web. The
122 fundamental take-homes are that nodes vary in their resolution, edges vary in what kind
123 of process they represent and the intersection of these, defined by meta- vs. realised webs,
124 underpins distinct lines of inquiry and constraints on the type of inference we can make
125 with networks. Here we reveal five core constraints across evolutionary and ecological
126 scales that further delineate the transition from meta- to realised webs, exposing processes
127 that determine the nature of links among nodes: evolutionary compatibility, co-occurrence,

128 abundance, diet choice, and non-trophic interactions Figure 1.

129 [Figure 1 about here.]

130 3.1 Processes that determine the feasibility of an interaction

131 Evolutionary compatibility and co-occurrence are the two principal processes that defines
132 the feasibility of an interaction between two species. The scale of inference and set
133 of processes embodied in these two constraints typically combine to define a ‘list’ of
134 interactions that are viable/feasible and defined strictly as present/absent. Reflecting on
135 the previous section, nodes are typically species and rules defining edges are defined by
136 trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight
137 into each process.

138 Evolutionary compatibility

139 This constraint is defined by shared (co)evolutionary history between consumers and
140 resources [52–55] which is manifested as ‘trait complementarity’ between two species [56].
141 In this body of theory, the consumer has the ‘correct’ set of traits that allow it to acquire
142 and consume the resource. Interactions that are not compatible are defined as forbidden
143 links [50]; *i.e.*, they are not physically possible and will *always* be absent within a network.
144 Networks do not properly arise from models based on this constraint. Instead, interacting
145 species pairs are defined and these are represented as binary (possible vs forbidden) or
146 probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are
147 quantified as the confidence of a specific interaction being *possible* between two species. A
148 network constructed based on evolutionary compatibility is conceptually aligned with a
149 ‘global metaweb’, and gives us information as to the global feasibility of links between
150 species pairs despite the fact that they do not co-occur (see Figure 1).

151 (Co)occurrence

152 The co-occurrence of species in both time and space is a fundamental requirement for an
153 interaction between two species to occur (at least in terms of feeding links). Although
154 co-occurrence data alone is insufficient for building an accurate and ecologically meaningful
155 representation of *feeding links* [58], it is still a critical process that determines the possible
156 realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially
157 constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1
158 this would be the metawebs for regions one and two.

159 We reinforce that these two constraints don't deliver a network *per se*, but a list of feasible
160 species pairs. Although it is possible to build a network from the list of interactions
161 generated by these constraints, it is important to be aware that the structure of this
162 network is not constrained by any community context - just because species are able to
163 interact does not mean that they will [60,61].

164 3.2 Processes that realise networks

165 In contrast to the above, here we highlight three processes that influence the *realisation* of
166 an interaction between species and thus form the conceptual basis for realised networks.
167 As we show in Figure 1, a 'truly realised' network is the product of properties of the
168 community (**abundance** and **non-trophic interactions**) and the individual (**diet**
169 **choice**). This represents a conceptual shift from considering the feasibility for species
170 pairwise interactions to considering the edge as a representation of energy flow. Such
171 a transition requires information about how the community, the environment and the
172 individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

173 Abundance

174 Abundance as a realising process emerges from a null model for energy acquisition:
175 organisms feeding randomly will consume resources in proportion to their abundance
176 [63]. Here, abundance of different prey species influences the distribution of links in a
177 network [64] by defining a preference linked to individuals among species meeting [47,60].
178 Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can
179 define the distribution and strength of links. Of note, however, is that such abundance
180 constrained interactions are not necessarily contingent on there being any compatibility
181 between species [65–67].

182 Diet choice

183 It is well established that consumers make more active decisions than eating items in
184 proportion to their abundance [63]. Ultimately, consumer choice is underpinned by
185 an energetic cost-benefit framework centred around profitability and defined by traits
186 associated with acquisition and consumption of a resource [68,69]. Energetic constraints
187 are invoked to construct networks in a myriad of ways [*e.g.*, 42,70–72].

188 Unlike metaweb approaches, these models generate realised webs as emergent outcomes of
189 consumer behaviour. We also here make a distinction, developed below, with models like

the Niche Model [39], where diet choice is implicit in its probabilistic network generating function, but it is working to replicate the *expected* structure of the network, and this structure does not emerge from node-based rules. Note that we select diet choice as a term to capture rules linked to optimal foraging [73] and metabolic theory [74] for capturing the energetic constraints on the distribution and strength of interactions.

Non-trophic interactions

We include non-trophic interactions [see 75] here not as a determinant of links, but a modifier of them - they are the community context above and beyond co-occurrence and abundance. Non-trophic interactions include competition for space, predator interference, refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or decrease mortality. These interactions specifically modify either the realisation or strength of trophic interactions [28,31,76–78] and represent direct (*e.g.*, predator *a* outcompetes predator *b*) and indirect (*e.g.*, mutualistic/facilitative interactions) mechanisms.

Some interactions, such as pollination, occupy an intermediate position in this framework, as they combine trophic components (*e.g.*, resource consumption) with non-trophic effects that influence reproduction, recruitment, and population persistence [79–81]. They operate on the realisation of a network by altering the fine-scale distribution and abundance of species and relative contributions of direct and indirect effects to biomass, persistence, stability, and the functioning of the communities [75,82–84].

4 Network construction

The above five processes are central to understanding the assumptions inherent in building different types of networks. Each of the processes, or combinations thereof, deliver a unique set of boundary conditions on what a network represents and can be used for. Here we build on the introduction of these five processes to further categorise the approaches to constructing networks. The utility of a network depends on the processes used to construct it. Different construction approaches encode different assumptions about how interactions arise, and therefore support different ecological questions.

4.1 Why construct networks?

Networks are a representation of biodiversity. In a perfect world, we might know about all interactions. However, the empirical collection of interaction data is both costly and

challenging to execute [50,85,86]. In the absence of robust empirical data, we use ‘models’ that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.*, 87,88–90], predict the feasibility of interaction among pairs of species, or directly predict network structure [see 91 for a broader discussion].

Networks are unique in delivering more than just estimates of species richness. As noted in the introduction, a network embodies the organising structure of biodiversity and allows numerous opportunities for ‘downstream’ analysis, including the comparison of structures, estimation of energy flux or extinction dynamics and ultimately form the structural inputs to dynamical systems models that facilitate ecological and conservation relevant inference about productivity-diversity-stability-function relationships [27] in space and time. But making such inferences requires careful attention to one or more of the processes discussed in Section 3. While these network representations may be simplifications of the truth [92], they remain critically useful tools as they allow us to move beyond simple descriptions of species richness to test hypotheses about community architecture and ecosystem functioning that would be otherwise impossible to assess.

4.2 Construction through induction

Constructing feasible or realised networks can be framed as an ‘inductive reasoning’ process where insight and generalisation arise from a set of observations and relationships around feeding. Inductive reasoning as a foundation for network construction is implemented at node and network levels.

4.2.1 Species specific induction

When applied at the node level, species-specific networks are created based on expected feeding interactions. When applied at the network level, networks are judged by their structural properties. All methods in this inference space rest on a set of three assumptions: there are a set of ‘feeding rules’ that underpin interaction feasibility [93]; these rules are phylogenetically conserved [54,94]; and they can be specified by matching the traits between consumer and resource.

Evolutionary compatibility and co-occurrence constraints have been critical to the construction of ‘first draft’ networks for communities for which we have no interaction data [57]. They are also central to interpolation in data poor regions and predicting interactions for ‘unobservable’ communities *e.g.*, prehistoric networks [43,95,96] or future, novel community

assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions among species relative to their distribution by accounting for the role of the environment and the role of species interactions [98–101].

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching [102,103], while mechanistic approaches often rely on trait-based relationships between consumers and resources [104,105]. Alternatively, statistical and machine-learning methods infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny [106].

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.*, traits) using some sort of binary classifier (see [106] for an overview). These include generalised linear models [*e.g.*, 107], random forest [*e.g.*, 108], trait-based k-NN [*e.g.*, 109], and Bayesian models [*e.g.*, 110,111].

Finally, graph embedding uses the structural features of a known network to infer the position of species in an unknown network through the decomposition of the interaction onto the embedding space (see [51]). This decomposition relies on a combination of ecological proxies (*e.g.*, phylogenetic relatedness [57]) in conjunction with known interactions to infer the latent values of species, which can then be mapped onto decomposition of a known network.

4.2.2 Structure-based induction

These models generate ecologically realistic network structures using simple probabilistic rules. They are commonly used as null expectations, for comparative analyses, and as inputs to dynamical models. The determination of links between species is not directly linked to properties of the nodes. This means these networks are usually not species specific. Although they require little empirical information, they encode explicit assumptions about expected network structure.

Stochastic network models use a probabilistic ruleset about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [112,113] and that network structure can be determined by distributing interactions along single dimension (the ‘niche axis’; [114]). Typically, these models parametrise some aspect of the network structure (although see [112] for a parameter-free model). These models include the most used network generator, the Niche model [39], as well as the original

283 Cascade model [5] and the derived Nested hierarchy model [115]. Even though these
 284 networks are derived without any real-world data they are still able to recover the structure
 285 of empirical networks [116]. These models often form the basis for dynamic models *e.g.*,
 286 the allometric trophic network [24,26] and bioenergetic food web models [34].

287 **4.3 Construction through deduction**

288 In contrast to metaweb construction, realised networks require assumptions about the
 289 ecological processes that determine whether feasible interactions are expressed. These
 290 approaches are deductive because interactions emerge from explicit assumptions about
 291 encounters, foraging behaviour, energetic constraints, or interaction modification. These
 292 approaches operationalise the abundance, diet-choice, and non-trophic processes introduced
 293 in Section 3.1. The resulting networks are widely used to study energy flux, extinction
 294 dynamics, and ecosystem functioning. They also provide the structural backbone for
 295 dynamical systems modelling to address questions about stability-structure-productivity-
 296 function relationships, secondary extinction dynamics, species invasion and climate change.
 297 There are two broad groups of models in this deductive category.

298 These models capture the behaviour of the nodes by explicitly considering the properties
 299 of the different species in the community. Which means that there is a degree of variance
 300 in which links are predicted between species unlike the more ‘static’ predictions made by
 301 inductive models. However, these networks are costly to construct in real world settings
 302 (requiring data about the entire community, as it is the behaviour of the system that
 303 determines the behaviour of the part) and also lack the larger diet niche context afforded
 304 by metawebs.

305 **4.3.1 Abundance-based models**

306 Neutral networks are built on the assumption that foraging decisions are tied *only* to the
 307 abundance of species within the community [117,118]. Here links are solely determined by
 308 the relative abundance of the different species in the community. Although unrealistic as
 309 a complete explanation, neutral models can be combined with inductive approaches to
 310 generate more localised predictions [67].

311 **4.3.2 Abundance-based models**

312 There is a broader group of models that focus on determining interactions in terms of
 313 energetic constraints on diet breadth, often using the ratio of consumer-resource body

size as a proxy for capturing the energetic constraints of feeding. Models such as those developed by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as ‘realised’ owing to the energetic constraint placed on links. A different subset of diet models [*e.g.*, 42,119] use a diet choice approach, however like the stochastic network models they also embed assumptions on network structure. Thus, these models predict both interactions and network structure simultaneously, although they would benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility) parameterisation [35].

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/population specific measures of the effects of stress and disturbance to community level metrics of these impacts. These metrics, such as resilience and more generally stability, ecosystem function and biodiversity, are natural properties of networks. This suggests that modern conservation, risk assessment and resource management requires robust network tools to support decision making.

This is also true in the disciplines of ecology and environmental science and their focus on abundance, distribution, functions and services that biodiversity provides [4]. Major questions remain, for example, about stability-diversity-productivity relationships, the impacts of extinctions and invasions, and the impacts of multiple stressors operating at multiple ecological scales. A network approach to answering these types of questions specifically allows us to evaluate how environmental gradients and anthropogenic stress map through direct and indirect effects among species in a complex community and reveal fundamental patterns and understanding of processes in the natural world.

To effectively use networks to aid us in answering questions about conservation/risk assessment/management and core ecological theory, we need to be mindful that we are mapping the *correct* network representation to the question of interest [21]. Notably, there are certain questions that cannot be answered using specific network representations as the scale of the question of interest is fundamentally misaligned with the process captured

346 by a specific network representation (Section 3.1), the underlying data that is used to
 347 construct it (Section 4), or both.

348 Here we discuss and map the different network representations shown in Figure 1 to
 349 ‘appropriate’ research questions and agendas (see also Table 1). We also highlight some of
 350 the key methodological challenges that currently limit our conceptualisation of a ‘network’
 351 and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally, we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Network		
Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to <i>e.g.</i> , extinction, turnover, invasion and rewilding $\checkmark$	[120]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources Δ	[121]; [122]; [123]
	Rewiring capacity/potential of species by looking at the <i>entire</i> diets of species Δ	[124]; [31]; [125]; [126]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $\times$	[60]; [127]
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ	[101]; [128]; [129]

Network		
Representation	Example Research Question	Representative Studies
Realised Webs	Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey ✓	[130]; [131]
	The allocation of multiple stressors across networks Δ	[132]; [133]
	Temperature threshold to community collapse Δ	[134]; [135]
	Extinction and persistence after harvesting/invasion/extinction ✓	[136]; [137]
	Stability-diversity-productivity-function ✓	[138]; [121]
	Explicitly tying ecosystem level processes and nutrient flows to networks $\times$	[139]
	Meta communities and the idea of meta-network-communities Δ	[140]; [141]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change [142], species invasions, reintroductions, and rewilding, and the potential rewiring capacity [126]. Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

361 5.1.2 Regional Metawebs

362 Regional metawebs extend these questions by explicitly incorporating species co-occurrence
363 and therefore provide a more management-relevant perspective. They can be used to
364 refine species distribution models and projections of future community composition [29,40].
365 However, caution is required when comparing regional metaweb structure across space
366 or environmental gradients, as observed differences may reflect species turnover (*e.g.*,
367 β -diversity) rather than changes in interaction processes.

368 5.1.3 Realised networks

369 Realised networks are best suited to questions concerning how community and environmen-
370 tal context shape interaction structure and ecosystem functioning [143]. They provide the
371 appropriate framework for studying stability, resilience, biodiversity dynamics, ecosystem
372 functioning, extinction cascades, invasions, climate change impacts, and network rewiring
373 through time. By capturing interactions that are actually occurring, realised networks
374 allow investigation of how perturbations propagate through communities and influence
375 persistence.

376 The increasing availability of long-term interaction datasets is expanding opportunities
377 to address these questions [27,144]. However, empirical datasets often accumulate in-
378 teractions across extended periods, potentially obscuring temporal variation in realised
379 interactions [10]. Developing approaches that better reconcile empirical networks with
380 realised community dynamics therefore remains an important challenge.

381 5.2 Key methodological challenges

382 As noted above, the three network types highlight longstanding methodological challenges
383 that limit both the precision and accuracy of ecological inference. Here we briefly review
384 these challenges and emerging opportunities to address them.

385 **Understanding what empirical data represents:** Robust inference requires under-
386 standing what constitutes an observed interaction, whether recorded directly (predation
387 events) or indirectly (*e.g.*, gut contents or stable isotopes). Because empirical networks
388 often accumulate observations across space and time, they may be conceptually closer to
389 metawebs than realised networks.

390 **The validation of network structure:** Considerable progress has been made in
391 assessing how well models recover pairwise interactions [91,145], yet there remains no clear

framework for evaluating their ability to recover network structure [114,146]. This raises two related questions: what constitutes an appropriate benchmark, and which aspects of network recovery matter most? For metawebs, accurately identifying both present and forbidden links may be essential, whereas for realised webs it remains unclear whether recovering pairwise interactions, aggregate properties (*e.g.*, connectance), or both should be the primary objective.

Transitioning between metawebs and realised webs: Most approaches for modelling realised networks do not explicitly incorporate evolutionary constraints (although see [147]; [68]). Progress will likely require either ensemble approaches [148,149] or methods for downscaling metawebs into realised networks [e.g., 150]. However, any such transition must retain clarity about the meaning of links - structurally realistic networks do not necessarily represent realised prey choice or energy flow.

Developing frameworks that allow transitioning between metawebs and realised representations will also facilitate integration with metacommunity, metaecosystem, and ecosystem-level theory [151,152]. Doing so requires clear definitions of what constitutes a network and where its spatial and temporal boundaries lie [153].

Making networks more tractable in applied spaces: The application of networks to conservation and management remains limited by difficulties in defining network boundaries, aligning them with management units, and interpreting network metrics in policy-relevant ways [154]. Addressing these challenges will require stronger links between network structure and ecosystem function, alongside careful matching of analytical tools and network representations to management objectives [128,155].

Taken together, these challenges highlight three overarching messages. (i) Network representations are inseparable from the data and assumptions used to construct them; (ii) validation and benchmarking must be aligned with the intended network type and research question; and (iii) greater conceptual clarity is needed when transitioning between metawebs and realised networks, particularly in applied contexts. Explicitly articulating these distinctions is essential if networks are to be used rigorously and transparently across scales.

6 Concluding Remarks

1. **Network representations encode different processes:** Different network representations capture distinct assumptions, data structures, and ecological processes; understanding the interplay between a network’s structure and the processes it encodes is critical for determining which questions it can address.
2. **Network representations are not interchangeable:** There is no universally optimal network representation. The utility of a network depends entirely on how well its underlying assumptions and data align with the intended scale and process of analysis.
3. **Explicit assumptions are essential for evaluation:** Explicitly articulating the assumptions behind network assembly facilitates a more critical evaluation of network suitability and inferential power. It is important that we move beyond the uncritical and default reliance on conventional network representations in favour of understanding the consequences of selecting a specific representation or reconstruction/modelling approach.
4. **A unified framework improves application and progress:** The framework presented here provides a structured basis for comparing network representations and evaluating their suitability across conceptual, methodological, and applied contexts. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

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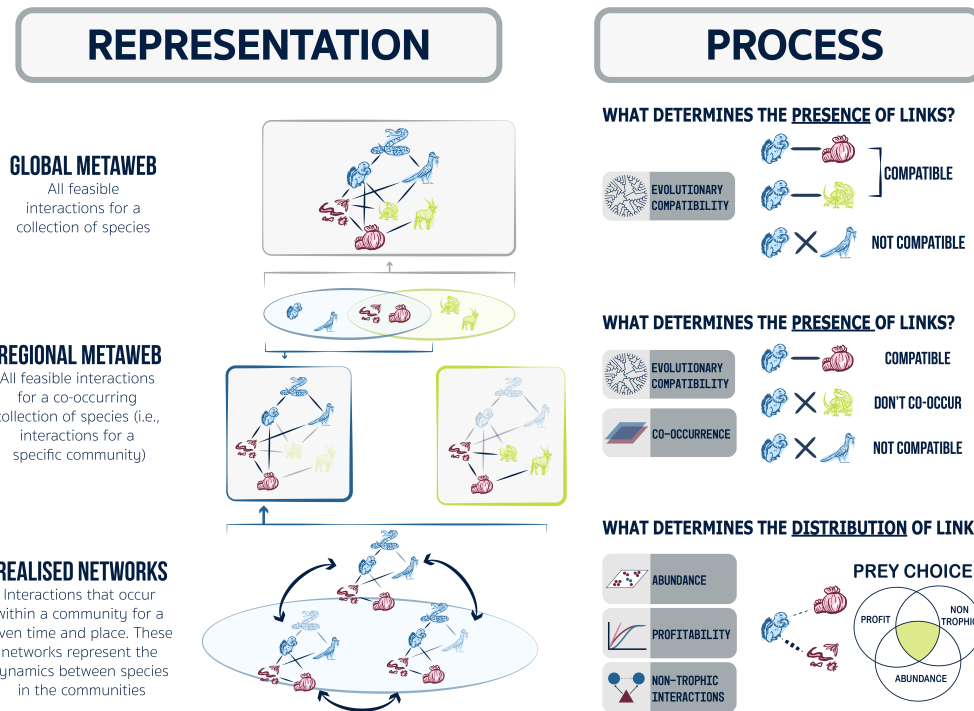


Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in specific spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.